FLUID POWER SYSTEMS (MEC 431)

Master Tutorial & Teaching Guide with Comprehensive Explanations

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Comprehensive Table of Contents:

1. **Module 1:** Introduction to Fluid Power — Physics, Components & Power Chains
2. **Module 2:** Fundamental Governing Laws — Pascal, Continuity & Orifice Equations
3. **Module 3:** Hydraulic Actuators & Circuit Dynamics — Standard vs. Differential Regeneration
4. **Module 4:** Fluid Properties & Hydraulic Oils (Chapter 2 In-Depth) — Viscosity, Density, Bulk Modulus, Thermal Expansion & Cavitation
5. **Module 5:** Fluid Friction & Line Transmission Dynamics — Hagen-Poiseuille, Resistance & Loss Analysis
6. **Module 6:** Valve Clearances, Internal Leakage & Viscous Friction
7. **Module 7:** Structural Compliance & Cylinder Lost Motion (Oil + Wall Elasticity)
8. **Module 8:** ISO 1219 / ANSI Y32.10 Graphical Symbols Master Reference
9. **Module 9:** Master Unit Conversions Table & Fast Calculation Formulas
10. **Module 10:** Fully Solved Exercise Sheet 1 with Detailed Pedagogical Explanations
11. **Module 11:** Fully Solved Exercise Sheet 2 with Comprehensive Analysis
12. **Module 12:** Fully Solved Exercise Sheet 3 (All 9 Problems) with Deep Theoretical Notes

Module 1: Introduction to Fluid Power — Physics, Components & Power Chains

1.1 Physical Essence of Fluid Power

Fluid power is the technology that deals with the generation, control, and transmission of power using pressurized fluids. Unlike mechanical transmissions (gears, cams, belts) that are constrained by rigid geometric pathways, fluids have *no fixed shape* and can transmit forces around corners, through flexible hoses, and across long distances with virtually zero backlash.

> **Hydrostatics vs. Hydrodynamics:**
>
> - **Hydrostatic (Fluid Power) Systems:** Power is transmitted primarily by the **pressure energy** of a fluid confined in an enclosed boundary (e.g., hydraulic jacks, presses, cylinders). Flow velocities are moderate ($v \approx 2 - 6\ \mathrm{m/s}$), while pressure is extremely high ($P \approx 50 - 350\ \mathrm{bar}$). Kinetic energy is negligible compared to static pressure energy.
> - **Hydrodynamic (Fluid Flow) Systems:** Power is transmitted primarily by the **kinetic energy** of the fluid moving at high velocities (e.g., Pelton wheels, torque converters, centrifugal pumps). Pressures are relatively low.

1.2 The Power Transmission Chain

The transmission chain in any industrial hydraulic machinery proceeds as follows:

Electric Motor / Engine $\xrightarrow{\text{Mechanical Energy: } T \cdot \omega}$ **Hydraulic Pump** $\xrightarrow{\text{Fluid Energy: } P \cdot Q}$ **Valves & Piping**
$\xrightarrow{\text{Controlled Flow \& Pressure}}$ **Actuator (Cylinder/Motor)** $\xrightarrow{\text{Mechanical Work: } F \cdot v \text{ or } T \cdot \omega}$ **External Load**

1.3 The 6 Essential Elements of a Complete Circuit

Element	Physical Function	Key Engineering Parameters
1. Reservoir (Tank)	Stores oil, dissipates heat, allows de-aeration, settles impurities.	Capacity = 3 to 5 times pump delivery ($V \approx 3 - 5 \times Q_P$).
2. Hydraulic Pump	Positive displacement unit; creates continuous flow against resistance.	Displacement V_d [cm³/rev], volumetric efficiency η_v, max pressure rating.
3. Pressure Relief Valve	Normally-closed safety valve; opens when pressure reaches setpoint to protect components.	Cracking pressure, full-flow pressure, response time.
4. Directional Control Valve (DCV)	Directs fluid flow to extend, retract, or hold the actuator.	Number of ports/positions (e.g., 4/3), center bypass configuration.
5. Actuator (Cylinder/Hydraulic Motor)	Converts hydraulic power back into linear or rotary mechanical power.	Bore diameter D, rod diameter d, stroke length, pressure rating.
6. Conditioning Equipment	Filters (suction, pressure, return), oil coolers, heaters, breathers.	Filtration rating (e.g., $10\ \mu\mathrm{m}\beta$-ratio), heat dissipation capacity.

> 💡 **TA Teaching Tip for Whiteboard Presentation:**
> Remind students: "*Pumps do NOT create pressure; pumps create flow! Pressure is generated ONLY by the resistance to flow*

> *imposed by the load and fluid friction.*" Ask students: If a pump discharges into an open bucket with no nozzle, what is the gauge pressure at the pump outlet? (Answer: Zero!). Pressure only rises when you restrict the flow or apply a load to a piston.

Module 2: Fundamental Governing Laws — Pascal, Continuity & Orifice Equations

2.1 Pascal's Law (Microscopic View & System Implications)

Formulated by Blaise Pascal in 1653, it states: "*Pressure applied to a confined liquid is transmitted undiminished in all directions throughout the liquid and acts with equal force on equal areas and at right angles to them.*"

$$P = \frac{F_1}{A_1} = \frac{F_2}{A_2} \implies F_2 = F_1 \times \left(\frac{A_2}{A_1}\right) = F_1 \times \left(\frac{D_2}{D_1}\right)^2$$

Conservation of Energy: Hydraulic systems multiply *force*, but they **cannot multiply energy**. If the small piston moves downward by distance s_1, the volume displaced is $V = A_1 s_1$. Since liquids are practically incompressible, this exact volume enters the large cylinder: $A_1 s_1 = A_2 s_2 \implies s_2 = s_1(A_1/A_2)$. Therefore, Work in = Work out ($F_1 s_1 = F_2 s_2$).

2.2 The Continuity Equation

Based on the Law of Conservation of Mass. For steady, incompressible flow ($\rho = \text{constant}$):

$$Q = A_1 v_1 = A_2 v_2 = \text{constant} \implies v_2 = v_1 \left(\frac{A_1}{A_2}\right) = v_1 \left(\frac{D_1}{D_2}\right)^2$$

When pipe diameter decreases, velocity increases quadratically. Higher velocity leads to dramatically increased turbulence and frictional pressure drops.

2.3 The Orifice Flow Equation

An orifice is a sudden restriction in a flow passage. Deriving from Bernoulli's equation between upstream section (1) and the vena contracta (2):

> $P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2 \implies v_2 = \sqrt{\frac{2(P_1 - P_2)}{\rho} + v_1^2}$

Assuming approach velocity $v_1 \ll v_2$ and introducing the discharge coefficient C_d (which accounts for jet contraction and frictional head losses):

$$\boxed{Q = C_d A_o \sqrt{\frac{2\Delta P}{\rho}}} \iff \Delta P = \frac{\rho Q^2}{2 C_d^2 A_o^2}$$

Orifice Geometry	Typical C_d Value	Behavioral Characteristic
Sharp-edged Orifice ($L/d < 0.5$)	$C_d \approx 0.611$	Discharge coefficient is **nearly independent of viscosity** and temperature! Highly stable in flow control valves.
Short-tube Orifice ($0.5 < L/d < 3$)	$0.13 \leq C_d \leq 0.816$	Dependent on Reynolds number and oil viscosity. Suffers from flow variations as oil heats up.

2.4 Hydraulic and Mechanical Power Relationships

Hydraulic Power: $N_h = P \cdot Q$ [W] (P in Pa, Q in m³/s)

Engineering Metric Units: $N_h\ [\mathrm{kW}] = \frac{P\ [\mathrm{bar}] \times Q\ [\mathrm{L/min}]}{600}$

Linear Mechanical Power: $N_m = F \cdot v$ [W] (F in N, v in m/s)

Rotary Mechanical Power: $N_m = T \cdot \omega = \frac{2\pi N T}{60}$ [W] (T in N·m, N in rpm)

Module 3: Hydraulic Actuators & Circuit Dynamics — Standard vs. Differential

3.1 Asymmetric Double-Acting Cylinder Geometry

A standard industrial double-acting cylinder has a single piston rod. This produces unequal areas:

- **Full Piston Area (Cap / Bore End):** $A_p = \frac{\pi}{4} D^2$ (Active during extension)
- **Piston Rod Cross-Sectional Area:** $A_r = \frac{\pi}{4} d^2$
- **Annulus (Rod End) Area:** $A_{an} = A_p - A_r = \frac{\pi}{4}(D^2 - d^2)$ (Active during retraction)

3.2 Standard Extension vs. Retraction Mechanics

Parameter	Standard Extension (Cap End Fed)	Standard Retraction (Rod End Fed)
Active Effective Area	$A_{eff} = A_p$	$A_{eff} = A_{an} = A_p - A_r$
Thrust / Pull Force	$F_{ext} = P \cdot A_p$ (Max force)	$F_{ret} = P \cdot A_{an}$ (Reduced force)
Piston Velocity	$v_{ext} = \frac{Q_P}{A_p}$ (Slower)	$v_{ret} = \frac{Q_P}{A_{an}}$ (Faster)
Displaced Flow to Tank	$Q_{out} = v_{ext} \cdot A_{an} = Q_P \left(1 - \frac{A_r}{A_p}\right) < Q_P$	$Q_{out} = v_{ret} \cdot A_p = Q_P \left(\frac{A_p}{A_{an}}\right) > Q_P$ ⚠️

> ⚠️ **Critical Engineering Warning for Sizing Return Lines:**
> Notice that during cylinder retraction, the flow leaving the cap end is $Q_{out} = Q_P \times (A_p / A_{an})$. For a typical ratio of $A_p / A_{an} = 2$, if the pump supplies 50 L/min, the flow escaping to the reservoir through the DCV is **100 L/min**! Sizing the directional valve or return lines based only on pump flow Q_P causes severe throttling, huge backpressure, and blown seal rings.

3.3 Differential (Regenerative) Connection Mechanics

In a differential circuit, both the cap end and rod end are connected simultaneously to system pressure P. Why does the piston extend rather than stall?

> Net force balance on the piston: $F_{net} = P \cdot A_p - P \cdot A_{an} = P(A_p - (A_p - A_r)) = P \cdot A_r$

Because $A_p > A_{an}$, the extension force dominates! Furthermore, the fluid displaced from the rod end cannot return to tank; it is routed directly back into the cap end to augment the pump flow:

$$Q_{in} = Q_P + Q_{ret} \implies v_{diff} \cdot A_p = Q_P + v_{diff} \cdot A_{an} \implies v_{diff}(A_p - A_{an}) = Q_P \implies \boxed{v_{diff} = \frac{Q_P}{A_r}}$$

> **The Hydraulic Gearbox Analogy:**
> Comparing differential extension with standard extension: $\frac{v_{diff}}{v_{ext}} = \frac{Q_P / A_r}{Q_P / A_p} = \frac{A_p}{A_r} > 1$, and $\frac{F_{diff}}{F_{ext}} = \frac{P \cdot A_r}{P \cdot A_p} = \frac{A_r}{A_p} < 1$.
> The speed increases by exactly the same factor that the force decreases: **Power remains identical!** Regenerative circuits act like shifting a car transmission into high gear—ideal for rapid approach strokes of stamping presses, drill feeds, and injection mold clampers.

Module 4: Fluid Properties & Hydraulic Oils (Chapter 2 In-Depth)

4.1 Viscosity: Dynamic vs. Kinematic

Viscosity is the measure of a fluid's internal resistance to shear flow. According to Newton's Law of Viscosity:

$$\tau = \mu \frac{du}{dy} \quad \Longrightarrow \quad \mu = \frac{\tau}{du/dy} \quad [\mathrm{N \cdot s/m^2 = Pa \cdot s}]$$

Kinematic viscosity ν is the ratio of dynamic viscosity to density:

$$\nu = \frac{\mu}{\rho} \quad [\mathrm{m^2/s}] \quad \text{where 1 Stokes (St)} = 10^{-4}\ \mathrm{m^2/s}, \quad 1\ \mathrm{cSt} = 10^{-6}\ \mathrm{m^2/s} = 1\ \mathrm{mm^2/s}$$

ISO Viscosity Grade (VG): Industrial hydraulic oils are designated by their kinematic viscosity at a standardized reference temperature of **40°C**. For example, *ISO VG 46* indicates an oil with $\nu = 46\ \mathrm{cSt} \pm 10\%$ at 40°C.

4.2 Viscosity Index (VI) & Pressure Dependency

- **Viscosity Index (VI):** An empirical unitless scale indicating the effect of temperature changes on viscosity. A high VI (e.g., VI > 150 for synthetic or multi-grade aircraft fluids) indicates minimal viscosity thinning over wide temperature swings. A low VI (e.g., VI < 90) indicates rapid loss of film thickness as oil heats up.
- **Effect of Pressure:** While temperature decreases viscosity exponentially, pressure increases viscosity. At pressures typical of standard industrial hydraulics (< 350 bar), the viscosity increase is moderate. However, above 700 bar (ultra-high pressure hydraulics), viscosity rises dramatically.

4.3 Compressibility & Bulk Modulus (B)

Liquids are not strictly incompressible. When subjected to huge hydraulic pressures, oil molecules are forced closer together. The Bulk Modulus of Elasticity B measures this stiffness:

$$B = -\frac{\Delta P}{\Delta V / V_o} \quad \Longrightarrow \quad \frac{\Delta V}{V_o} = -\frac{\Delta P}{B}$$

For standard mineral hydraulic oils, $B \approx 1.2 \text{ to } 1.8\ \mathrm{GPa}$ ($1.2 - 1.8 \times 10^9\ \mathrm{N/m^2}$). Entrained air bubbles drastically reduce effective bulk modulus, making the system "spongy", reducing hydraulic natural frequency, and causing servo-valve instability.

4.4 Thermal Expansion & Trapped Fluid Hazards

When the temperature of hydraulic oil changes by ΔT, its unrestrained volumetric expansion is:

$$\Delta V_T = V_o \cdot \alpha \cdot \Delta T \quad \text{where for mineral oil, } \alpha \approx 0.0007\ \mathrm{K^{-1}}$$

If fluid is trapped inside a completely rigid container with closed valves, zero net volume change is possible ($\Delta V = \Delta V_T + \Delta V_P = 0$):

$$V_o \alpha \Delta T - \frac{V_o \Delta P}{B} = 0 \implies \boxed{\Delta P = B \cdot \alpha \cdot \Delta T}$$

> ⚠️ **Catastrophic Safety Implication:**
> For typical oil ($B = 1.4\ \mathrm{GPa},\ \alpha = 0.0007\ \mathrm{K^{-1}}$), a temperature rise of just **10°C** creates a pressure spike of:

$\Delta P = (1.4 \times 10^9) \times 0.0007 \times 10 = 9.8\ \text{MPa} = 98\ \text{bar}$!
If a cylinder or pipe section is isolated by closed ball valves and exposed to sunlight or machine heat, pressure escalates to hundreds of bars, rupturing pipes or cracking valve bodies. *Every trapped line must be protected by a thermal relief valve!*

4.5 Vapor Pressure & The Cavitation Phenomenon

The Physics of Cavitation:

1. **Low Pressure Zone Formation:** When local static pressure drops below the oil's **saturated vapor pressure** (or when dissolved air separates at negative gauge pressure), vapor and air cavities spontaneously form.
2. **Migration:** These bubbles are carried by the flow into high-pressure regions (e.g., inside the pump rotating group or past a valve seat).
3. **Violent Implosion:** Surrounded by high pressure, the bubbles collapse inward asymmetrically. As the bubble walls crash together, liquid micro-jets impact nearby metal surfaces at velocities over 1000 m/s and local micro-pressures exceeding **10,000 bar (1 GPa)**!
4. **Consequences:** Severe pitting/fatigue erosion of bronze valve plates, intense screaming noise, high-frequency vibration, and rapid thermal cracking of the oil.

Module 5: Fluid Friction & Line Transmission Dynamics

5.1 Laminar Flow in Pipelines: Hagen-Poiseuille Derivation

Consider fully developed laminar flow of fluid through a straight horizontal pipe of inner diameter D, radius R, and length L.

Step 1: Force Balance on a Fluid Cylinder:
Consider a cylindrical fluid element of radius r and length L. Pressure forces on the two ends must balance the viscous shear force on the circumference:
$(P_1 - P_2) \cdot \pi r^2 = \tau(r) \cdot (2\pi r L) \implies \tau(r) = \frac{\Delta P}{2L} r$

Step 2: Velocity Profile from Newton's Law of Viscosity:
Since velocity decreases with radius ($du/dr < 0$), $\tau = -\mu \frac{du}{dr}$:
$-\mu \frac{du}{dr} = \frac{\Delta P}{2L} r \implies du = -\frac{\Delta P}{2\mu L} r \, dr$
Integrating from r to wall R with no-slip boundary condition ($u(R) = 0$):
$u(r) = \frac{\Delta P}{4\mu L}(R^2 - r^2)$ — Parabolic Velocity Profile!

Step 3: Integrating for Volumetric Flow Rate Q:
$Q = \int_0^R u(r) \cdot 2\pi r \, dr = \frac{2\pi \Delta P}{4\mu L} \int_0^R (R^2 r - r^3) dr = \frac{\pi \Delta P}{2\mu L} \left[\frac{R^4}{2} - \frac{R^4}{4} \right] = \frac{\pi R^4 \Delta P}{8\mu L}$
Substituting $R = D/2$ ($R^4 = D^4/16$):
$Q = \frac{\pi D^4 \Delta P}{128 \mu L} \implies \boxed{\Delta P = \frac{128 \mu L Q}{\pi D^4} = R_{hyd} \cdot Q}$

5.2 Hydraulic Resistance & Electric Circuit Analogy

The term $R_{hyd} = \frac{128 \mu L}{\pi D^4}$ is the **linear hydraulic resistance** of the transmission line. It is perfectly analogous to electric resistance $V = I \cdot R$:

$$\text{Pressure Drop: } \Delta P = R_{hyd} \cdot Q \iff \text{Voltage Drop: } V = R_{el} \cdot I$$
$$\text{Power Loss: } \Delta N = \Delta P \cdot Q = R_{hyd} \cdot Q^2 \iff \text{Joule Heating: } P = R_{el} \cdot I^2$$

The Power-of-Four Sensitivity ($1/D^4$):
Notice that hydraulic resistance is inversely proportional to the **fourth power of internal diameter**. If an engineer replaces a 20 mm pipe with a 10 mm pipe (halving the diameter):
$\left(\frac{D_{old}}{D_{new}} \right)^4 = \left(\frac{20}{10} \right)^4 = 2^4 = 16$ times higher pressure loss!
Even a tiny deposit, internal corrosion, or choosing an undersized tube causes massive power dissipation that turns directly into destructive system heat.

5.3 Turbulent Flow & Darcy-Weisbach Formulation

Flow regime is dictated by the Reynolds number: $Re = \frac{\rho v D}{\mu} = \frac{v D}{\nu} = \frac{4 \rho Q}{\pi \mu D}$.

- **Laminar Regime:** $Re < 2000$. Use Hagen-Poiseuille: $\lambda = \frac{64}{Re}$.
- **Transition Regime:** $2000 \le Re \le 4000$. Unstable, erratic pressure drops.
- **Turbulent Regime:** $Re > 4000$. Darcy-Weisbach equation: $\Delta P = \lambda \frac{L}{D} \frac{\rho v^2}{2}$.
  For smooth hydraulic steel tubing, Blasius equation yields: $\lambda = \frac{0.3164}{Re^{0.25}}$.

Module 6: Valve Clearances, Internal Leakage & Viscous Friction

6.1 Internal Leakage through Concentric Annular Clearances

In spool valves, pumps, and motors, moving mechanical parts cannot have contact seals due to excessive friction and wear. Instead, minute radial clearances c (2 to 10 μm) separate the moving spool and the fixed sleeve.

Treating the thin annular clearance as an unwrapped parallel plate channel of width $W = \pi D$, gap height c, and length L:

Laminar flow between stationary flat plates with gap c:

$$Q_L = \frac{Wc^3}{12\mu L}\Delta P \implies \boxed{Q_L = \frac{\pi D c^3}{12\mu L}\Delta P}$$

Leakage Resistance: $R_L = \frac{12\mu L}{\pi D c^3} \implies \Delta P = R_L \cdot Q_L$

Power Dissipated by Leakage: $\Delta N = Q_L \cdot \Delta P = R_L \cdot Q_L^2 = \frac{\pi D c^3}{12\mu L}\Delta P^2$

⚠️ **The Cubic Law of Clearance Wear (c^3):**
Because leakage scales with c^3, doubling the radial clearance due to abrasive contaminant wear increases internal leakage by: $(2)^3 = 800\%$ $(8 \text{ times higher!})$
This explains why slightly worn pumps and valves suddenly lose volumetric efficiency and overheat the oil tank without any visible external oil drip.

6.2 Viscous Friction & Damping Coefficient on Spool Lands

When a spool with n lands moves axially at velocity v inside a sleeve filled with oil of viscosity μ and radial gap c:

Assuming linear velocity gradient in the micro-gap: $\frac{du}{dy} = \frac{v}{c}$
Shear stress on spool outer surface: $\tau = \mu\frac{v}{c}$
Total shear surface area: $A_{shear} = \pi D \sum_{i=1}^{n} L_i$
Viscous drag force: $F_v = \tau \cdot A_{shear} = \mu\frac{v}{c}(\pi D \sum L_i) = f_v \cdot v$

Viscous Damping Coefficient: $\boxed{f_v = \frac{\mu\pi D \sum L_i}{c}}$ $[\text{N} \cdot \text{s/m}]$

Module 7: Structural Compliance & Cylinder Lost Motion

7.1 Combined Effective Bulk Modulus (B_{eff})

When pressure surges in a hydraulic cylinder and its connecting pipes, two elastic phenomena happen simultaneously:

1. The hydraulic oil compresses (ΔV_{oil}).
2. The cylindrical steel walls expand radially due to hoop stress (ΔV_{wall}).

The total system compliance is the sum of oil compressibility and container wall elasticity:

$$\frac{1}{B_{eff}} = \frac{1}{B_{oil}} + \frac{1}{B_{container}} = \frac{1}{B_{oil}} + \frac{D}{2 \cdot t_{wall} \cdot E_{steel}}$$

Where: D = Internal diameter, t_{wall} = Wall thickness, E_{steel} = Young's Modulus of wall material ($\approx 210\ \text{GPa}$).

7.2 Piston Lost Motion (Shift) Calculation

When an external force increment ΔF strikes a locked cylinder, pressure rises by $\Delta P = \frac{\Delta F}{A_p}$. The fluid in both the cylinder body and the supply line compresses while the walls stretch:

> Total elastic volume change: $\Delta V_{total} = \Delta V_{cyl} + \Delta V_{pipe} = \frac{V_{cyl} \cdot \Delta P}{B_{eff,cyl}} + \frac{V_{pipe} \cdot \Delta P}{B_{eff,pipe}}$
>
> Physical piston displacement ("Lost Motion"): $\boxed{\Delta x = \frac{\Delta V_{total}}{A_p}}$

Module 8: ISO 1219 / ANSI Y32.10 Graphical Symbols Master Reference

8.1 Hydraulic Lines & Connections

Symbol Line Style	Standard ISO Meaning	Application in Circuit Diagrams
Solid Thick Line (——)	Working / Main Line	Carries high-pressure pump supply, return flow to tank, cylinder lines.
Dashed Line (- - -)	Pilot / Control Line	Carries signal fluid to actuate remote valves, pilot-operated relief, sequence.
Dotted Line (····)	Drain Line	Carries internal leakage from motor/valve casing back to tank at 0 bar.
Circle with Dots / Chain	Component Enclosure	Groups multiple valves inside a single manifold block.

8.2 Pumps, Actuators & Energy Converters

Symbol Geometry	Meaning	Symbol Geometry	Meaning
Circle + Solid triangle pointing **out**	Fixed Displacement Hydraulic Pump	Circle + Solid triangle pointing **in**	Fixed Displacement Hydraulic Motor
Circle + Open triangle pointing out	Pneumatic Compressor	Circle + Open triangle pointing in	Pneumatic Air Motor
Circle + Triangle + 45° arrow	Variable Displacement Pump/Motor	Rectangle + Rod on one side	Single-Rod Double-Acting Cylinder

8.3 Directional Control Valves (DCV) Naming & Center Conditions

Named as: **Ports / Positions** (e.g., 4/3 DCV = 4 flow ports, 3 spool positions).

Center Condition (4/3)	Port P	Port T	Ports A & B	Operational Effect on Machine
Closed Center	Blocked	Blocked	Blocked	Piston locked rigidly. Pump deadheads (must dump over relief).
Tandem Center	Connected to T	Connected to P	Blocked	Pump unloads to tank at 0 bar (saves power). Load remains locked!
Open Center	Connected to T	Connected to P	Connected to T	Pump unloads to tank. Cylinder floats freely without resistance.
Float Center	Blocked	Connected to A, B	Connected to T	P blocked (maintains line pressure). Actuator floats freely.

Module 9: Master Unit Conversions Table & Fast Calculation Formulas

Physical Quantity	SI Unit	Common Engineering Unit	Conversion Factor
Pressure (P)	Pascal (Pa = N/m²)	bar	$1\ \text{bar} = 10^5\ \text{Pa} = 0.1\ \text{MPa} = 100\ \text{kPa}$
		psi (lbf/in²)	$1\ \text{bar} \approx 14.5038\ \text{psi} \quad \vert \quad 1\ \text{psi} \approx 6894.76\ \text{Pa}$
Flow Rate (Q)	m³/s	L/min (LPM)	$1\ \text{L/min} = \frac{1}{60{,}000}\ \text{m}^3/\text{s} = 1.667 \times 10^{-5}\ \text{m}^3/\text{s}$
		GPM (US)	$1\ \text{GPM} \approx 3.78541\ \text{L/min} \approx 6.309 \times 10^{-5}\ \text{m}^3/\text{s}$
Kinematic Visc. (ν)	m²/s	Centistokes (cSt)	$1\ \text{cSt} = 1\ \text{mm}^2/\text{s} = 10^{-6}\ \text{m}^2/\text{s} = 10^{-2}\ \text{St}$
Dynamic Visc. (μ)	N·s/m² (Pa·s)	Poise (P) / Centipoise (cP)	$1\ \text{P} = 0.1\ \text{N} \cdot \text{s/m}^2 \quad \vert \quad 1\ \text{cP} = 10^{-3}\ \text{N} \cdot \text{s/m}^2$
Bulk Modulus (B)	Pa	GPa	$1\ \text{GPa} = 10^9\ \text{Pa} = 10{,}000\ \text{bar}$
Power (N)	Watt (W)	Kilowatt (kW) / Horsepower	$1\ \text{kW} = 1000\ \text{W} \quad \vert \quad 1\ \text{hp (metric)} \approx 735.5\ \text{W} \approx 0.736\ \text{kW}$

Module 10: Fully Solved Exercise Sheet 1 with Detailed Pedagogical Explanations

Problem 1: Simple Hydraulic Jack (Pascal's Principle)

Problem Statement: A simple hydraulic jack has a pump piston diameter $d = 20\ \text{mm}$ and a load ram diameter $D = 100\ \text{mm}$. A weight of $F_2 = 2500\ \text{N}$ rests on the ram. Neglecting friction, calculate the hydraulic pressure in the system and the force F_1 that must be applied to the pump piston.

> **Physical Concept:** Both pistons are connected to the same continuous body of liquid. Under static equilibrium, Pascal's law guarantees that pressure is identical at both piston surfaces (neglecting small elevation head).

Step 1: Convert units to SI:
$d = 20\ \text{mm} = 0.02\ \text{m}, \quad D = 100\ \text{mm} = 0.1\ \text{m}$

Step 2: Calculate cross-sectional areas:
$A_2 = \frac{\pi}{4}D^2 = \frac{\pi}{4}(0.1)^2 = 7.854 \times 10^{-3}\ \text{m}^2$
$A_1 = \frac{\pi}{4}d^2 = \frac{\pi}{4}(0.02)^2 = 3.1416 \times 10^{-4}\ \text{m}^2$

Step 3: Calculate required fluid pressure:
$P = \frac{F_2}{A_2} = \frac{2500}{7.854 \times 10^{-3}} = 318,310\ \text{Pa} = 318.31\ \text{kPa} \approx 3.183\ \text{bar}$

Step 4: Calculate input force on small piston:
$F_1 = P \cdot A_1 = 318,310 \times 3.1416 \times 10^{-4} = 100.0\ \text{N}$

Final Answers:

System Pressure: **P = 318.31 kPa (3.18 bar)**

Effort Force: **F_1 = 100 N**

> 💡 **TA Check Method:** Point out to students that $F_1 = F_2 \times (d/D)^2 = 2500 \times (20/100)^2 = 2500/25 = 100\ \text{N}$. The mechanical advantage of the hydraulic system is exactly 25.

Problem 2: Hand-Operated Jack with Lever & Stroke Counting

Problem Statement: A hand-operated hydraulic jack has a pump piston diameter of 1.5 in and a load piston diameter of 4 in. The handle is a Class 2 lever with a mechanical advantage of 8:2 (effort arm = 8 in, load arm = 2 in from pivot). The operator applies an effort of 30 lbf. The pump stroke is 3 in.
(a) Determine the maximum load force the jack can raise.
(b) How many complete pumping cycles are required to raise the load by 12 in?

(a) Force Calculation:
Lever moment balance around pivot: $F_{rod} \times 2 = F_{hand} \times 8 \implies F_{rod} = 30 \times \frac{8}{2} = 120\ \text{lbf}$
Pump piston area: $A_1 = \frac{\pi}{4}(1.5)^2 = 1.7671\ \text{in}^2$
Pressure generated: $P = \frac{F_{rod}}{A_1} = \frac{120}{1.7671} = 67.906\ \text{psi}$
Ram area: $A_2 = \frac{\pi}{4}(4)^2 = 12.5664\ \text{in}^2$
Load lifted: $F_{load} = P \cdot A_2 = 67.906 \times 12.5664 = 853.3\ \text{lbf}$

(b) Cycle Calculation:
Oil volume displaced per single pump stroke: $V_{stroke} = A_1 \times S_{pump} = 1.7671 \times 3 = 5.3014 \text{ in}^3$
Total oil volume needed to lift load 12 in: $V_{load} = A_2 \times S_{load} = 12.5664 \times 12 = 150.796 \text{ in}^3$
Number of cycles: $N = \frac{V_{load}}{V_{stroke}} = \frac{150.796}{5.3014} = 28.44 \implies \boxed{29 \text{ cycles}}$

Final Answers:

(a) Load Capacity: **F_load = 853.3 lbf** | (b) Pumping Cycles: **N = 29 complete cycles**

Problem 3: Pneumatic-to-Hydraulic Pressure Booster (Intensifier)

Problem Statement: An air-oil intensifier has an air piston area $A_{air} = 22 \text{ in}^2$ driven by shop air at $p_{air} = 110$ psi. The high-pressure hydraulic ram has an area $A_{oil} = 2 \text{ in}^2$. The boosted oil feeds a hydraulic press cylinder with area $A_{press} = 30 \text{ in}^2$. Find the oil pressure and total clamping force.

Force balance on the rigid connecting rod: $F = p_{air} \cdot A_{air} = p_{oil} \cdot A_{oil}$
$p_{oil} = p_{air} \times \left(\frac{A_{air}}{A_{oil}}\right) = 110 \times \left(\frac{22}{2}\right) = 110 \times 11 = 1210 \text{ psi}$
Clamping force: $F_{press} = p_{oil} \cdot A_{press} = 1210 \times 30 = 36,300 \text{ lbf}$

Final Answers:

Boosted Oil Pressure: **P_oil = 1210 psi (83.4 bar)** | Clamping Force: **F = 36,300 lbf (161.4 kN)**

Module 11: Fully Solved Exercise Sheet 2 with Comprehensive Analysis

Common Circuit Data (Problems 4 & 5):

A hydraulic cylinder operates under system pressure $P = 195 \text{ bar} = 195 \times 10^5 \text{ Pa}$, fed by pump flow $Q_P = 39 \text{ L/min} = 6.5 \times 10^{-4} \text{ m}^3/\text{s}$.

Piston bore diameter $D = 90 \text{ mm} = 0.09 \text{ m}$, Rod diameter $d = 50 \text{ mm} = 0.05 \text{ m}$.

Preliminary Area Computations:

Piston Area: $A_p = \frac{\pi}{4}(0.09)^2 = 6.3617 \times 10^{-3} \text{ m}^2$

Rod Area: $A_r = \frac{\pi}{4}(0.05)^2 = 1.9635 \times 10^{-3} \text{ m}^2$

Annular (Rod End) Area: $A_{an} = A_p - A_r = 4.3982 \times 10^{-3} \text{ m}^2$

Q4: Standard Extension Performance

1. Thrust Force: $F_{ext} = P \cdot A_p = (195 \times 10^5) \times (6.3617 \times 10^{-3}) = 124,053 \text{ N} = \mathbf{124.05 \text{ kN}}$

2. Forward Velocity: $v_{ext} = \frac{Q_P}{A_p} = \frac{6.5 \times 10^{-4}}{6.3617 \times 10^{-3}} = 0.10217 \text{ m/s} = \mathbf{10.22 \text{ cm/s}}$

3. Return Flow to Tank:

$Q_T = v_{ext} \cdot A_{an} = 0.10217 \times (4.3982 \times 10^{-3}) = 4.4937 \times 10^{-4} \text{ m}^3/\text{s} = \mathbf{26.96 \text{ L/min}}$

4. Hydraulic Power: $N_h = P \cdot Q_P = (195 \times 10^5) \times (6.5 \times 10^{-4}) = 12,675 \text{ W} = \mathbf{12.675 \text{ kW}}$

5. Mechanical Output: $N_m = F_{ext} \cdot v_{ext} = 124,053 \times 0.10217 = 12,675 \text{ W} = \mathbf{12.675 \text{ kW}}$

Q5: Differential (Regenerative) Connection Performance

1. Forward Velocity: $v_{diff} = \frac{Q_P}{A_r} = \frac{6.5 \times 10^{-4}}{1.9635 \times 10^{-3}} = 0.33104 \text{ m/s} = \mathbf{33.10 \text{ cm/s}}$

2. Cap-End Total Inflow:

$Q_{in} = v_{diff} \cdot A_p = 0.33104 \times (6.3617 \times 10^{-3}) = 2.106 \times 10^{-3} \text{ m}^3/\text{s} = \mathbf{126.36 \text{ L/min}}$

3. Regenerated Flow:

$Q_{regen} = v_{diff} \cdot A_{an} = 0.33104 \times (4.3982 \times 10^{-3}) = 1.456 \times 10^{-3} \text{ m}^3/\text{s} = \mathbf{87.36 \text{ L/min}}$

(Notice: $Q_{in} = Q_P + Q_{regen} = 39 + 87.36 = 126.36 \text{ L/min}$ — continuity holds!)

4. Thrust Force: $F_{diff} = P \cdot A_r = (195 \times 10^5) \times (1.9635 \times 10^{-3}) = 38,288 \text{ N} = \mathbf{38.29 \text{ kN}}$

5. Mechanical Output: $N_m = F_{diff} \cdot v_{diff} = 38,288 \times 0.33104 = 12,675 \text{ W} = \mathbf{12.675 \text{ kW}}$

Direct Engineering Comparison Table:

Performance Parameter	Standard Extension (Q4)	Differential Mode (Q5)	Relative Ratio
Extension Velocity (v)	10.22 cm/s	**33.10 cm/s**	**3.24× Faster** (A_p/A_r)
Available Thrust (F)	**124.05 kN**	38.29 kN	**3.24× Lower** (A_r/A_p)
Cap End Inflow (Q_{in})	39.0 L/min	**126.36 L/min**	3.24× Internal loop
Flow to Reservoir (Q_T)	26.96 L/min	**0.0 L/min** (Bypasses tank)	Tank loop eliminated
Net Power Consumed (N)	12.675 kW	12.675 kW	Identical (Ideal efficiency)

Module 12: Fully Solved Exercise Sheet 3 (All 9 Problems)

Problem 1: Pressure & Power Loss Expressions in Laminar Pipe Flow

Question: Derive an expression for the pressure and power losses in a hydraulic transmission line of constant diameter in the case of laminar flow.

Start from Darcy-Weisbach: $\Delta P = \lambda \frac{L}{D} \frac{\rho v^2}{2}$.
For laminar flow: $\lambda = \frac{64}{Re}$, where $Re = \frac{\rho v D}{\mu}$.
Substitute λ: $\Delta P = \left(\frac{64\mu}{\rho v D}\right) \frac{L}{D} \frac{\rho v^2}{2} = \frac{32 \mu L v}{D^2}$.
Substitute mean velocity $v = \frac{4Q}{\pi D^2}$:
$\boxed{\Delta P = \frac{128 \mu L Q}{\pi D^4} = R_{hyd} \cdot Q}$ where $R_{hyd} = \frac{128\mu L}{\pi D^4}$.
Power Loss: $\boxed{\Delta N = \Delta P \cdot Q = R_{hyd} \cdot Q^2 = \frac{128 \mu L}{\pi D^4} Q^2}$ [W].

Problem 2: Pressure Increment in Trapped Liquid under Temperature Rise

Question: Derive an expression for the pressure increment in a volume of liquid trapped in a rigid container when subjected to a temperature increase ΔT.

1. Unrestrained thermal volume expansion: $\Delta V_T = V_o \cdot \alpha \cdot \Delta T$.
2. Pressure compression from Bulk Modulus: $\Delta V_P = -\frac{V_o \Delta P}{B}$.
3. Rigid boundary constraint requires zero net change: $\Delta V_{total} = \Delta V_T + \Delta V_P = 0$.
$V_o \alpha \Delta T - \frac{V_o \Delta P}{B} = 0 \implies \boxed{\Delta P = B \cdot \alpha \cdot \Delta T}$.

Problem 3: Saturated Vapor Pressure & Cavitation in Hydraulic Systems

Question: Discuss briefly the effect of the saturated vapor pressure on the function of hydraulic systems.

When local fluid pressure drops below the oil's **saturated vapor pressure**, the oil boils locally at operating temperature, creating vapor-filled cavities (cavitation). As these cavities reach higher pressure zones, they collapse implosively, generating micro-jets of up to 1000 MPa. Consequences: surface pitting of pump gears/pistons, severe chatter noise, degraded volumetric efficiency, and rapid oxidation of the fluid.
Countermeasures: Increase pump suction line diameter, shorten inlet pipe, mount reservoir above pump level, or use a pressurized tank.

Problem 4: Percentage Variation in Volume under Pressure

Question: Calculate the percentage of variation in volume of liquid of bulk modulus $B = 1.2\ \text{GPa}$ if its pressure is increased by $12\ \text{MPa}$.

$B = -\frac{\Delta P}{\Delta V / V_o} \implies \left|\frac{\Delta V}{V_o}\right| = \frac{\Delta P}{B}$
$\frac{\Delta V}{V_o} = \frac{12 \times 10^6\ \text{Pa}}{1.2 \times 10^9\ \text{Pa}} = 0.010 = \mathbf{1.0\%}$

Percentage Volume Decrease: **ΔV/V = 1.0%**

Problem 5: Viscous Friction Coefficient for a Spool Valve

Question: Derive an expression for, and calculate, the viscous friction coefficient for the given spool valve: $L_1 = L_2 = 12$ mm, $D = 10$ mm, $c = 3$ μm, $\mu = 0.03\ \text{N}\cdot\text{s/m}^2$.

Shear stress: $\tau = \mu\frac{v}{c}$. Area of two cylindrical lands: $A_s = \pi D(L_1 + L_2)$.

Friction force: $F_v = \tau A_s = \left(\frac{\mu\pi D(L_1+L_2)}{c}\right)v = f_v \cdot v \implies \boxed{f_v = \frac{\mu\pi D(L_1 + L_2)}{c}}$.

Calculation:

$f_v = \frac{0.03\times\pi\times0.010\times(0.012+0.012)}{3\times10^{-6}} = \frac{0.03\times3.1416\times0.010\times0.024}{3\times10^{-6}} = \frac{2.2619\times10^{-5}}{3\times10^{-6}} = \mathbf{7.54\ N\cdot s/m}$.

Viscous Friction Coefficient: **f_v = 7.54 N·s/m**

Problem 6: Pressure Upstream of a Throttling Valve with Flow Regime Verification

Given: $Q = 15\ \text{L/min} = 2.5\times10^{-4}\ \text{m}^3/\text{s}$, $\rho = 860\ \text{kg/m}^3$, $\mu = 0.0185\ \text{N}\cdot\text{s/m}^2$,
$A_t = 2\ \text{mm}^2 = 2\times10^{-6}\ \text{m}^2$, $C_d = 0.611$.
Line 1: $D_1 = 15$ mm, $L_1 = 1.5$ m; Line 2: $D_2 = 10$ mm, $L_2 = 2.5$ m; Line 3: $D_3 = 12$ mm, $L_3 = 2.0$ m.

1. Reynolds Number Verification: $Re = \frac{4\rho Q}{\pi\mu D}$

Line 1: $Re_1 = \frac{4\times860\times2.5\times10^{-4}}{\pi\times0.0185\times0.015} = 987 < 2000$ (Laminar ✓)
Line 2: $Re_2 = \frac{0.86}{\pi\times0.0185\times0.010} = 1480 < 2000$ (Laminar ✓)
Line 3: $Re_3 = \frac{0.86}{\pi\times0.0185\times0.012} = 1233 < 2000$ (Laminar ✓)
Laminar assumption is verified for all lines.

2. Pipe Pressure Drops (Hagen-Poiseuille): $\Delta P = \frac{128\mu LQ}{\pi D^4}$

$\Delta P_1 = \frac{128\times0.0185\times1.5\times2.5\times10^{-4}}{\pi\times(0.015)^4} = 5,585\ \text{Pa} = \mathbf{5.59\ kPa}$
$\Delta P_2 = \frac{128\times0.0185\times2.5\times2.5\times10^{-4}}{\pi\times(0.010)^4} = 47,109\ \text{Pa} = \mathbf{47.11\ kPa}$
$\Delta P_3 = \frac{128\times0.0185\times2.0\times2.5\times10^{-4}}{\pi\times(0.012)^4} = 18,175\ \text{Pa} = \mathbf{18.18\ kPa}$

3. Orifice Pressure Drop Across Throttle:

$\Delta P_t = \frac{\rho Q^2}{2C_d^2A_t^2} = \frac{860\times(2.5\times10^{-4})^2}{2\times(0.611)^2\times(2\times10^{-6})^2} = \frac{5.375\times10^{-5}}{2.9866\times10^{-12}} = 17.997\times10^6\ \text{Pa} = \mathbf{17.997\ MPa}$

4. Pressure Upstream of Throttle:

Assuming layout: Pump → Line 1 → Line 2 → Throttle (A_t) → Line 3 → Tank (0 bar):

$P_{upstream} = \Delta P_1 + \Delta P_2 + \Delta P_t + \Delta P_3 = 5.59 + 47.11 + 17,997 + 18.18 = \mathbf{18,068\ kPa \approx 18.07\ MPa}$

Upstream Pressure: **P ≈ 18.07 MPa (180.7 bar)**

Problem 7: Pump Suction Line & Cavitation Analysis

Given: $Q = 59\ \text{L/min} = 9.833 \times 10^{-4}\ \text{m}^3/\text{s}$, Open tank ($P_{atm} = 101,325\ \text{Pa}$), Suction pipe $L = 1.5\ \text{m}$, $D = 14\ \text{mm} = 0.014\ \text{m}$, $\nu = 0.2\ \text{cm}^2/\text{s} = 2 \times 10^{-5}\ \text{m}^2/\text{s}$, $\rho = 870\ \text{kg/m}^3$.

1. Flow Velocity & Regime:

$v = \frac{Q}{\frac{\pi}{4}D^2} = \frac{9.833\times10^{-4}}{1.5394\times10^{-4}} = 6.388\ \text{m/s}$

$Re = \frac{vD}{\nu} = \frac{6.388\times0.014}{2\times10^{-5}} = 4472 > 4000 \implies \textbf{TurbulentFlow!}$

2. Friction Factor & Pressure Loss (Darcy-Weisbach):

Blasius: $\lambda = \frac{0.3164}{Re^{0.25}} = \frac{0.3164}{(4472)^{0.25}} = 0.0387$

$\Delta P = \lambda \frac{L}{D} \frac{\rho v^2}{2} = 0.0387 \times \left(\frac{1.5}{0.014}\right) \times \left(\frac{870\times(6.388)^2}{2}\right) = 0.0387 \times 107.14 \times 17,750 = \mathbf{73,600\ Pa}$

3. Pump Inlet Static Pressure:

$P_{inlet} = P_{atm} - \Delta P = 101,325 - 73,600 = \mathbf{27,725\ Pa\ (abs) = 0.277\ bar\ (abs)}$

Pump Inlet Pressure: **P_inlet = 27.7 kPa (abs) = -0.736 bar (gauge)**

⚠️ **Severe Cavitation Alert:** Absolute pressure at pump inlet is only 0.27 bar. Dissolved air will boil out violently! Suction line velocity should never exceed 1.2 m/s (here it is 6.39 m/s!). Pipe diameter must be enlarged to at least 32 mm.

Problem 8: Series Pipeline Friction Losses

Given: $Q = 10\ \text{L/min} = 1.667 \times 10^{-4}\ \text{m}^3/\text{s}$, $\rho = 850\ \text{kg/m}^3$, $\nu = 21\ \text{cSt} = 2.1 \times 10^{-5}\ \text{m}^2/\text{s}$, $L_1 = L_2 = 5\ \text{m}$, $D_1 = 15\ \text{mm}$, $D_2 = 10\ \text{mm}$.

$\mu = \rho\nu = 850 \times 2.1 \times 10^{-5} = 0.01785\ \text{N}\cdot\text{s/m}^2$.

Check Reynolds: $Re_1 = \frac{4\rho Q}{\pi\mu D_1} = 674 < 2000$ (Laminar); $Re_2 = \frac{4\rho Q}{\pi\mu D_2} = 1011 < 2000$ (Laminar).

$\Delta P_1 = \frac{128\mu L_1 Q}{\pi D_1^4} = \frac{128\times0.01785\times5\times1.667\times10^{-4}}{\pi\times(0.015)^4} = \mathbf{11,975\ Pa}$

$\Delta P_2 = \frac{128\mu L_2 Q}{\pi D_2^4} = \frac{128\times0.01785\times5\times1.667\times10^{-4}}{\pi\times(0.010)^4} = \mathbf{60,590\ Pa}$

$\Delta P_{total} = \Delta P_1 + \Delta P_2 = 11,975 + 60,590 = 72,565\ \text{Pa} = \mathbf{72.57\ kPa \approx 0.726\ bar}$

Total Pressure Loss: **ΔP_total = 72.57 kPa (0.726 bar)**

Problem 9: Piston Shift Due to Force Surge (Oil & Wall Compliance)

Given: $\Delta F = 105\ \text{kN} = 1.05 \times 10^5\ \text{N}$, $A_p = 0.0178\ \text{m}^2$, Pipe: $d = 15\ \text{mm}$, $h = 1\ \text{mm}$, $L = 6\ \text{m}$; Cylinder: $t = 5\ \text{mm}$, $x_o = 0.6\ \text{m}$; $B = 1.2\ \text{GPa}$, $E = 210\ \text{GPa}$.

1. Pressure Surge: $\Delta P = \frac{\Delta F}{A_p} = \frac{105,000}{0.0178} = 5.899 \times 10^6\ \text{Pa} = \mathbf{5.899\ MPa}$

2. Cylinder Bore: $D_{cyl} = \sqrt{\frac{4A_p}{\pi}} = \sqrt{\frac{4\times0.0178}{\pi}} = 0.1505\ \text{m} = 150.5\ \text{mm}$

3. Fluid Volumes:

$V_{cyl} = A_p \cdot x_o = 0.0178 \times 0.6 = 0.01068\ \text{m}^3$

$V_{pipe} = \frac{\pi}{4}d^2 L = \frac{\pi}{4}(0.015)^2 \times 6 = 1.060 \times 10^{-3}\ \text{m}^3$

4. Effective Bulk Moduli:

Pipe:

$$\frac{1}{B_{eff,p}} = \frac{1}{1.2\times10^9} + \frac{0.015}{2(0.001)(210\times10^9)} = 8.333\times10^{-10} + 3.571\times10^{-11} = 8.690\times10^{-10} \implies B_{eff,p} = 1.151\ \text{GPa}$$

Cylinder:

$$\frac{1}{B_{eff,c}} = \frac{1}{1.2\times10^9} + \frac{0.1505}{2(0.005)(210\times10^9)} = 8.333\times10^{-10} + 7.167\times10^{-11} = 9.050\times10^{-10} \implies B_{eff,c} = 1.105\ \text{GPa}$$

5. Volume Decreases & Piston Shift:

$$\Delta V_p = \frac{V_p \Delta P}{B_{eff,p}} = \frac{1.060\times10^{-3}\times5.899\times10^6}{1.151\times10^9} = 5.43\times10^{-6}\ \text{m}^3$$

$$\Delta V_c = \frac{V_c \Delta P}{B_{eff,c}} = \frac{0.01068\times5.899\times10^6}{1.105\times10^9} = 57.01\times10^{-6}\ \text{m}^3$$

$$\Delta V_{total} = \Delta V_p + \Delta V_c = 62.44\times10^{-6}\ \text{m}^3$$

Piston Shift: $\Delta x = \frac{\Delta V_{total}}{A_p} = \frac{62.44\times10^{-6}}{0.0178} = 3.508\times10^{-3}\ \text{m} = \mathbf{3.51\ mm}$

Total Piston Shift: **Δx = 3.51 mm**